Continuous coupling trajectories — run report

RUN

211549_cross-sweep_timing-calibration-post-in2-lv
Status: completed | Started: 2026-09-29T20:15:49.340992+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi1, phi2; sine: 0.1164 Hz; measured PAX rate: 6.91 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 59.35.
Recorded cycles: 4; bias values: phi1 target 1, phi2 target 1. Waveform tracking: correct within descriptive threshold. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): unresolved. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

End-to-end measurement-tool calibration after IN2 LV jumper repair: one high-signal bias per driven axis, explicit ASG zero phase, four recorded cycles, raw PAX transaction times retained, configured 0.120 s latency. Compare 0 and 120 ms offline.

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

Continuous coupling trajectories — run report

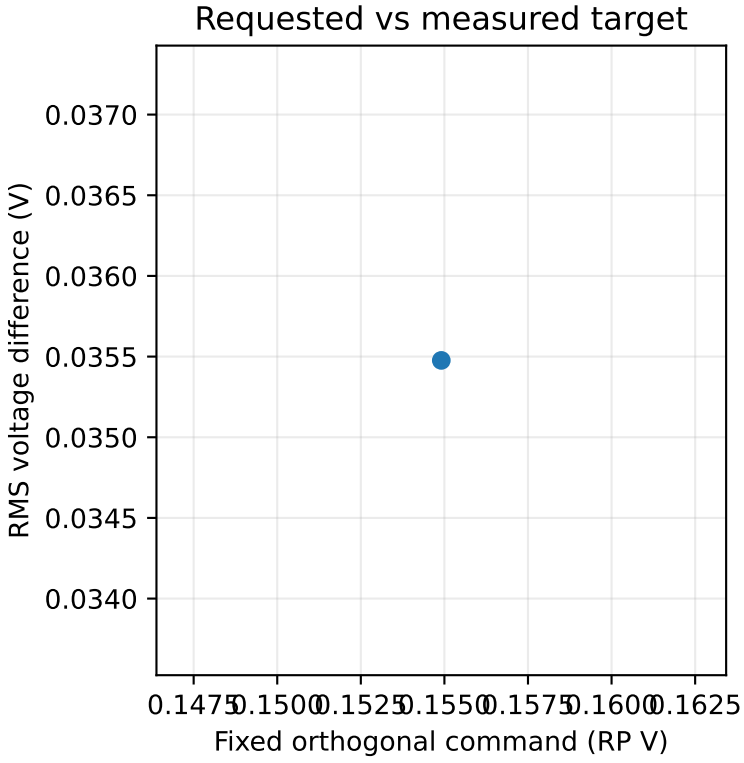
DATA

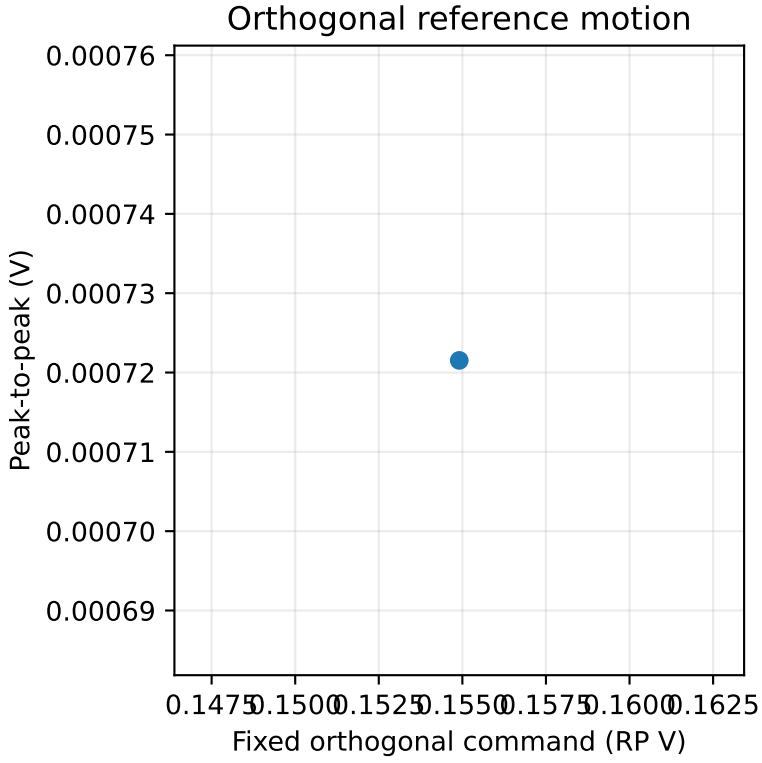
609 recorded samples; acquisition: 609
Source: data.csv; full acquisition settings: recipe.json; status: run.json

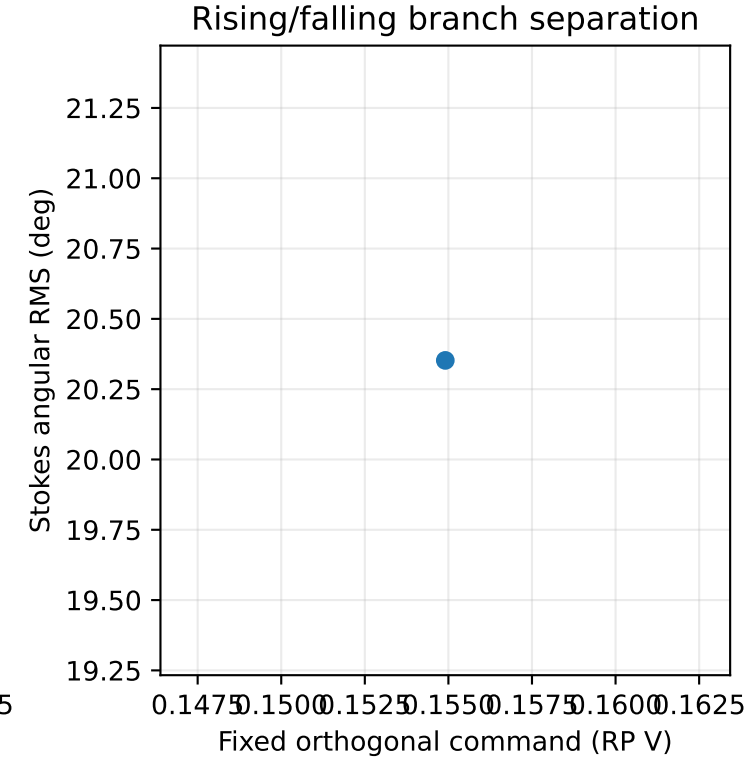
RAW DOP

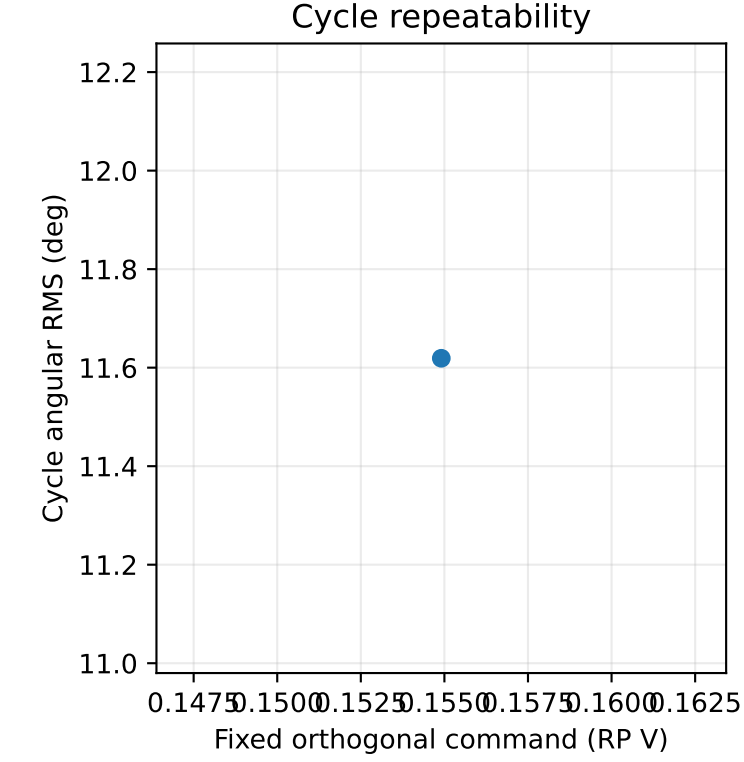
Mean 0.8611; range 0.6222 to 0.9209. 0 samples outside 0–1. Telemetry is not a new calibration.

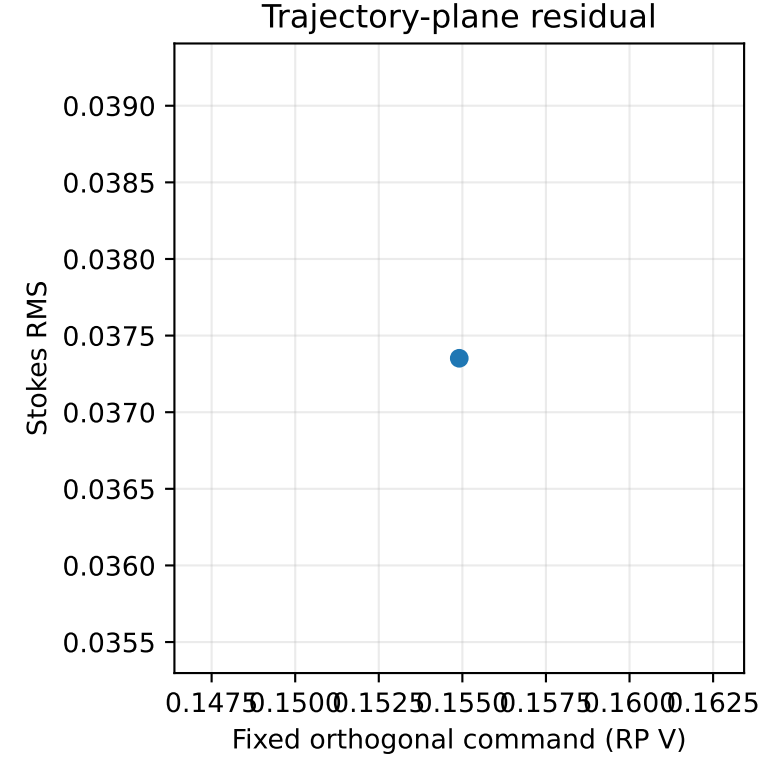
Continuous phi1 diagnostic — bias dependence

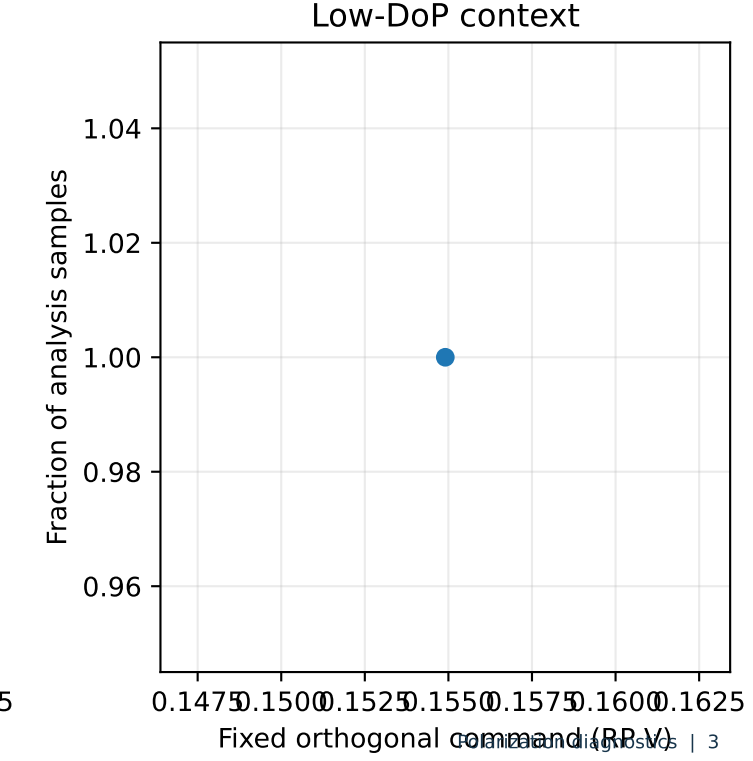




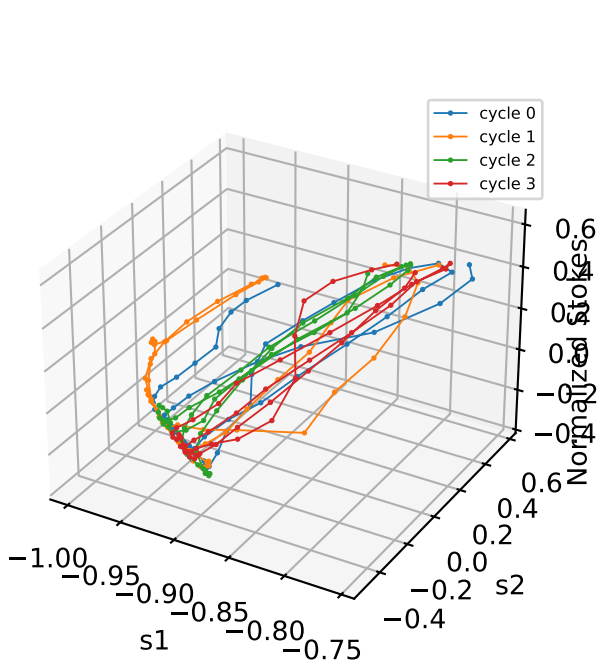


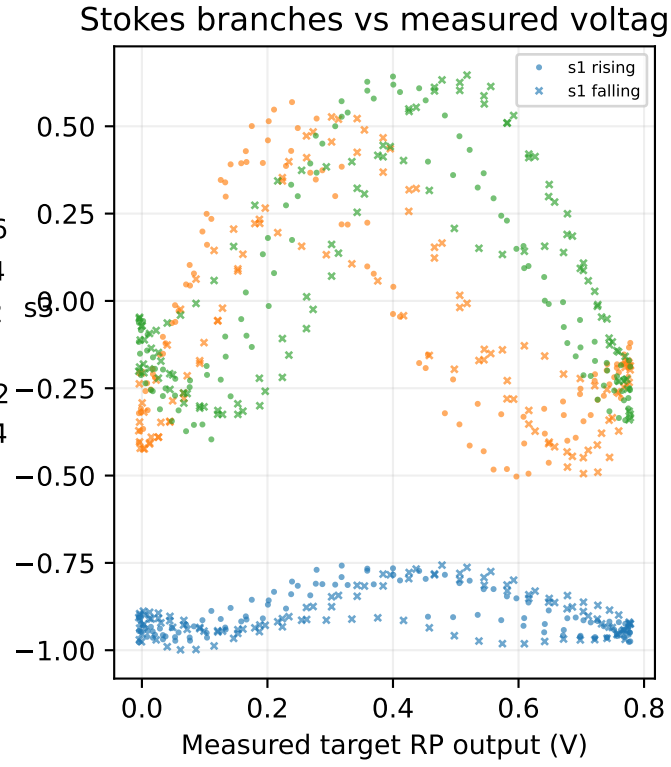


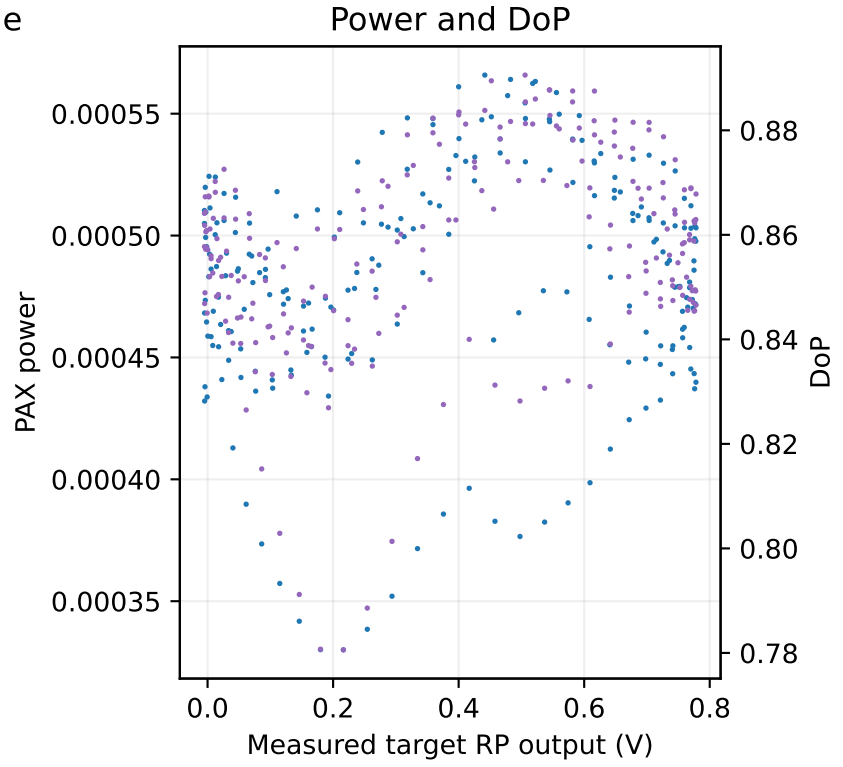


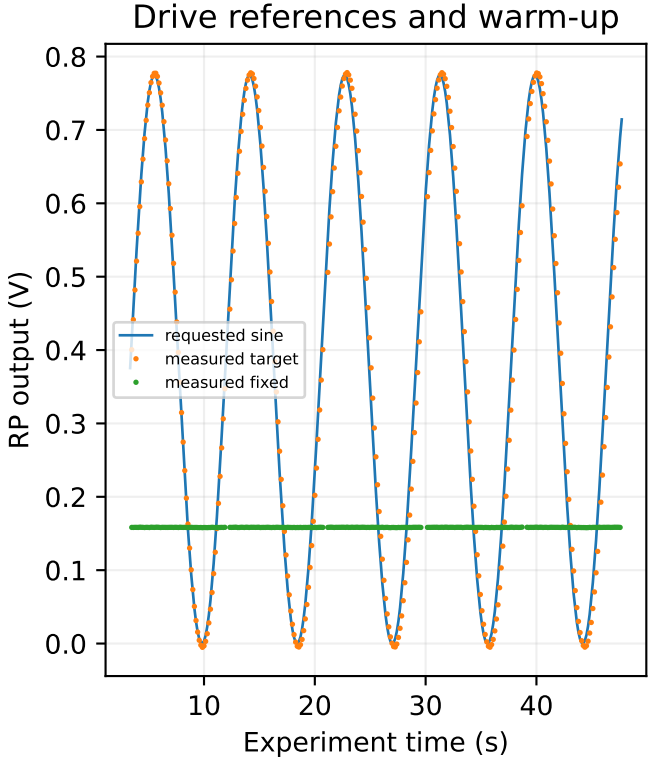


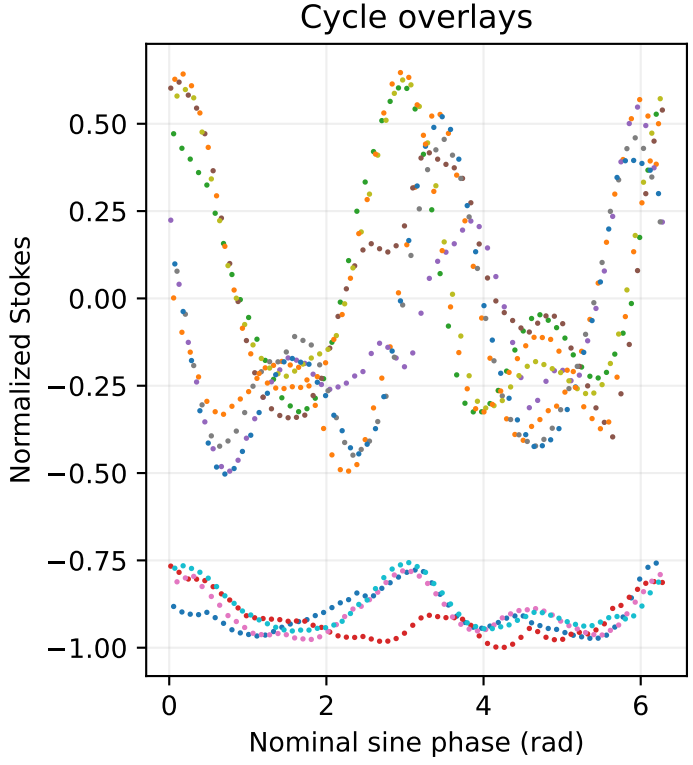
phi1 sine; fixed phi2 = 0.1549 RP V (segment 0)



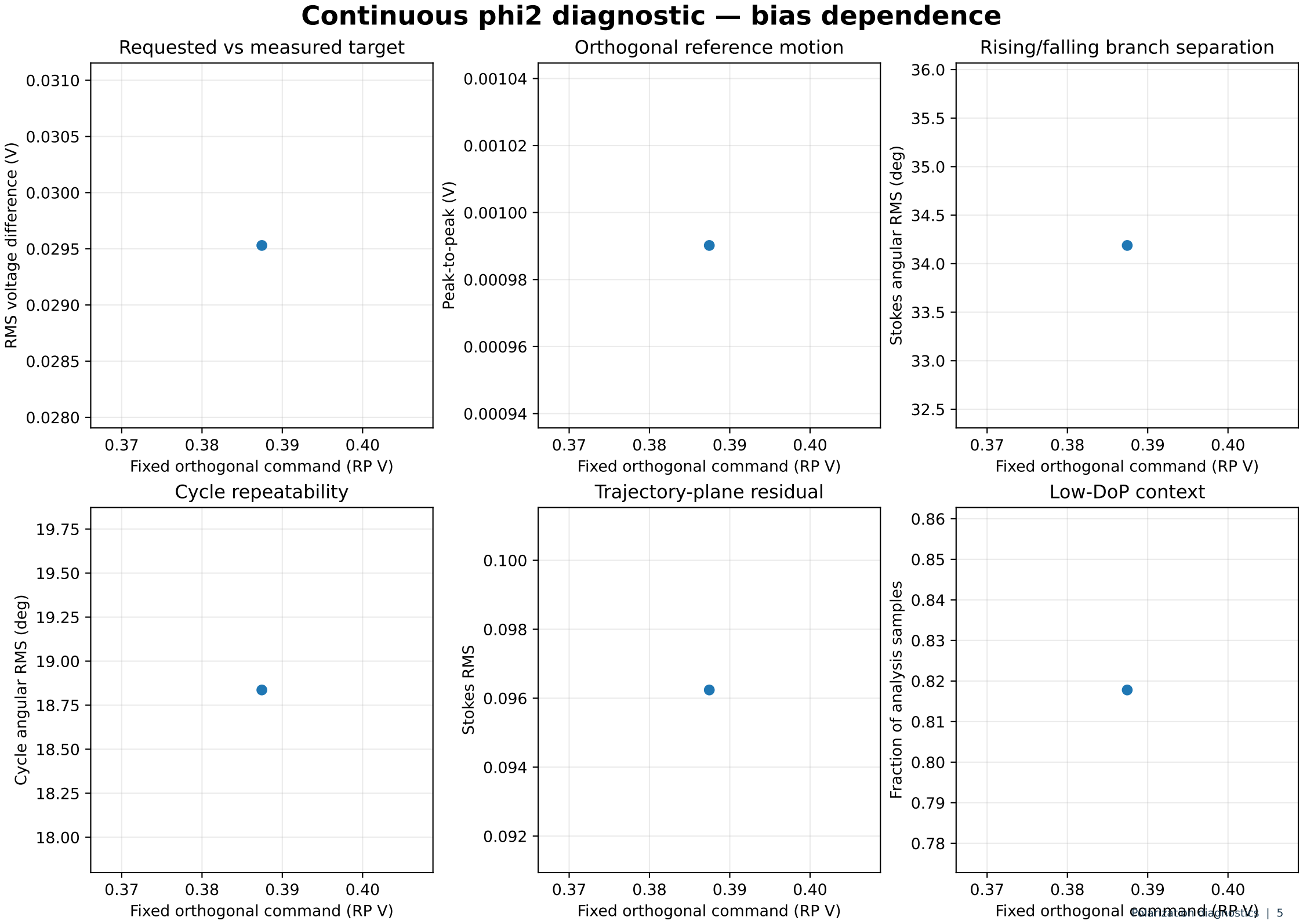




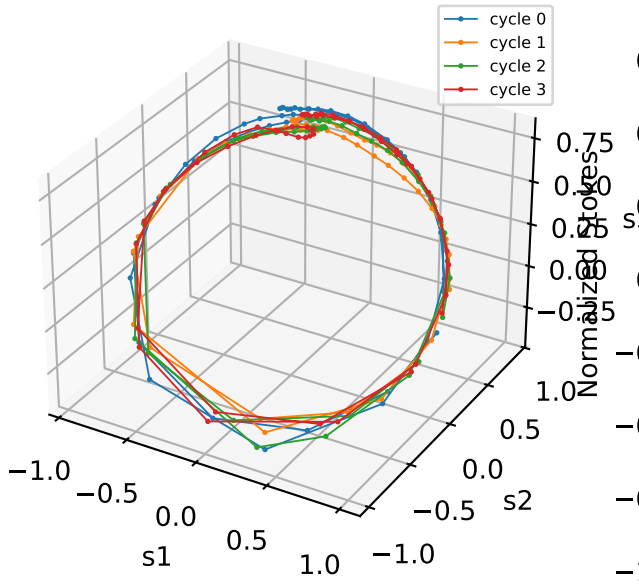




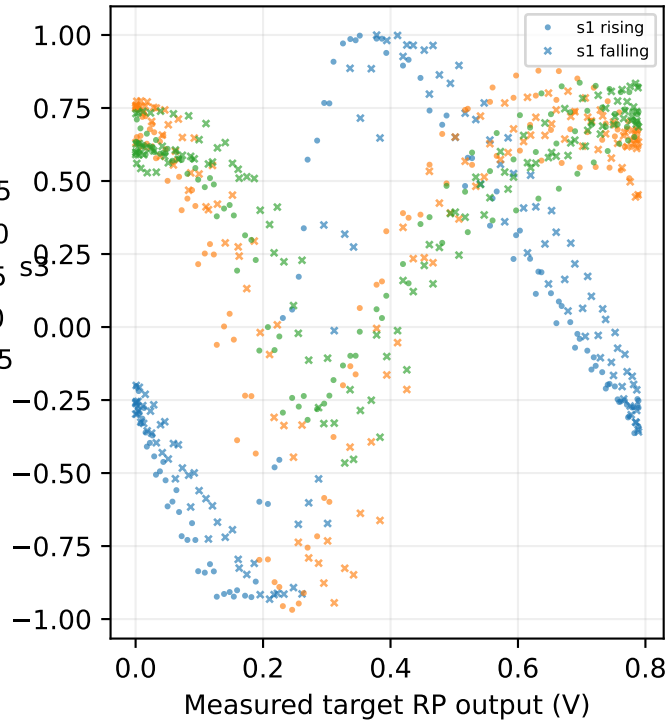
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.988, -0.024, -0.153]
Plane residual RMS: 0.03735
Rising/falling angular RMS: 20.4°
Rising/falling power RMS: 7.61e-05 PAX units
Cycle angular RMS: 11.6°
Cycle power RMS: 6.1e-05 PAX units
Cycle DoP RMS: 0.0202
Tracking RMS: 0.03548 V
Reference coverage: 96.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.06, -0.14], residual 0.0325
Cycle 1 plane: [0.99, -0.04, -0.10], residual 0.0503
Cycle 2 plane: [0.98, 0.01, -0.18], residual 0.0214
Cycle 3 plane: [0.98, -0.01, -0.18], residual 0.0164


Continuous phi2 diagnostic — bias dependence
Requested vs measured target
RMS voltage difference (V)
0.0310
0.0305
0.0300
0.0295
0.0290
0.0285
0.0280
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Orthogonal reference motion
Peak-to-peak (V)
0.00104
0.00102
0.00100
0.00098
0.00096
0.00094
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Rising/falling branch separation
Stokes angular RMS (deg)
36.0
35.5
35.0
34.5
34.0
33.5
33.0
32.5
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Cycle repeatability
Cycle angular RMS (deg)
19.75
19.50
19.25
19.00
18.75
18.50
18.25
18.00
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Trajectory-plane residual
Stokes RMS
0.100
0.098
0.096
0.094
0.092
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Low-DoP context
Fraction of analysis samples
0.86
0.85
0.84
0.83
0.82
0.81
0.80
0.79
0.78
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)

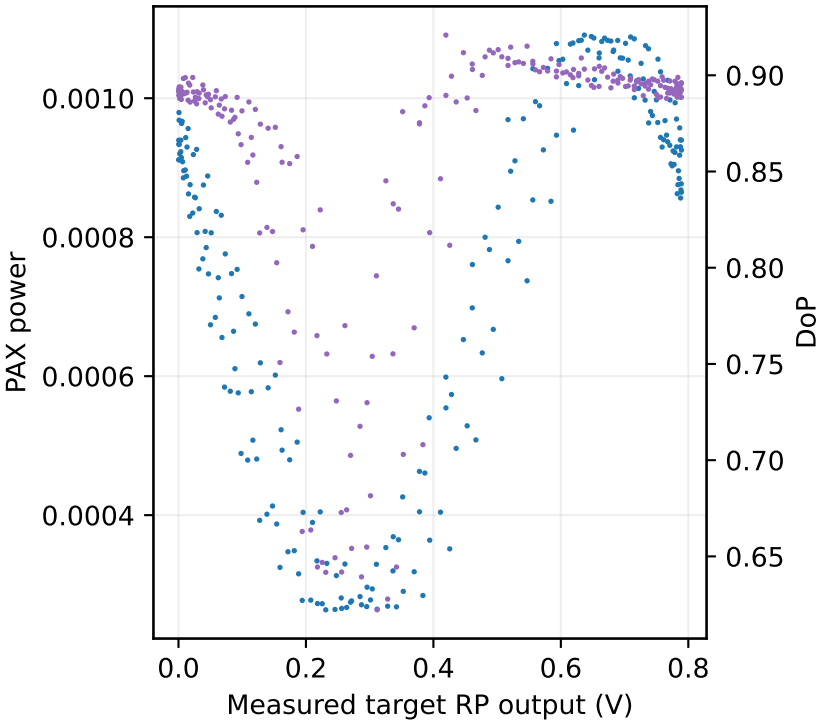
phi2 sine; fixed phi1 = 0.3875 RP V (segment 1)



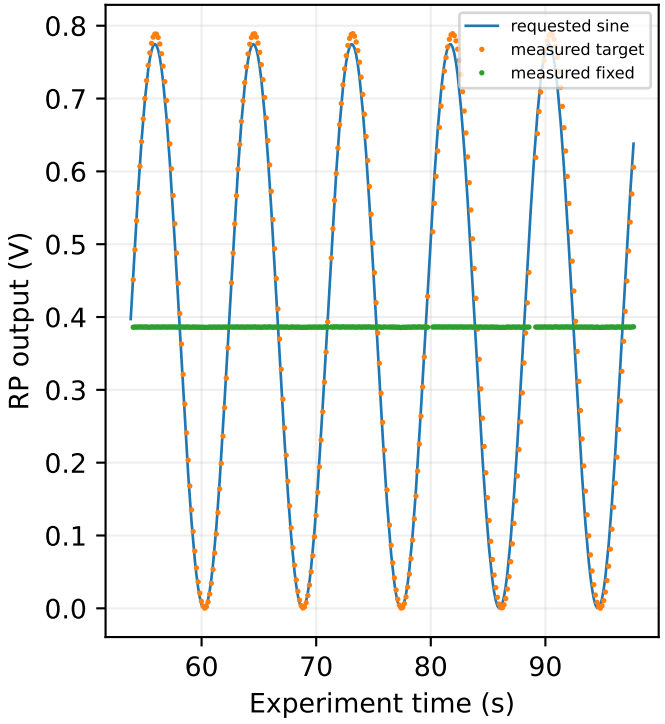
Stokes branches vs measured voltage



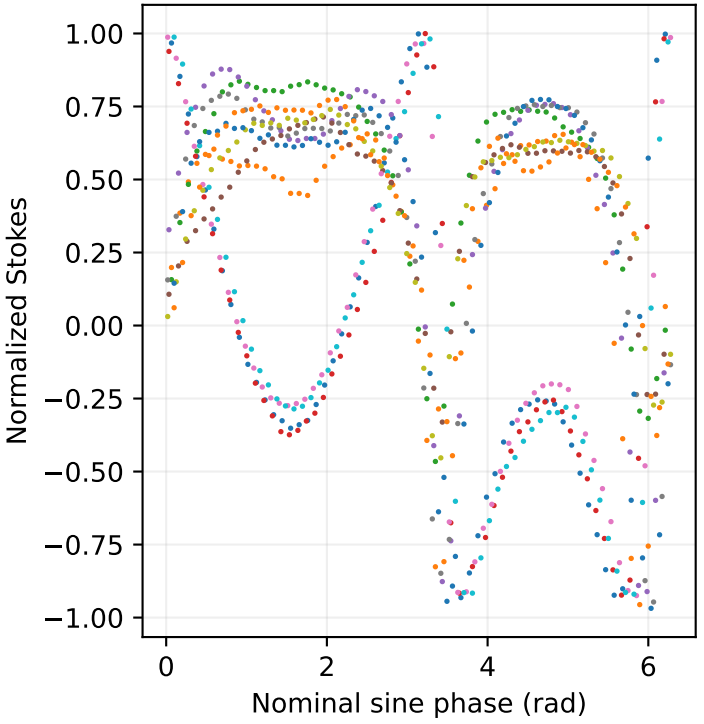
Power and DoP



Drive references and warm-up



Cycle overlays

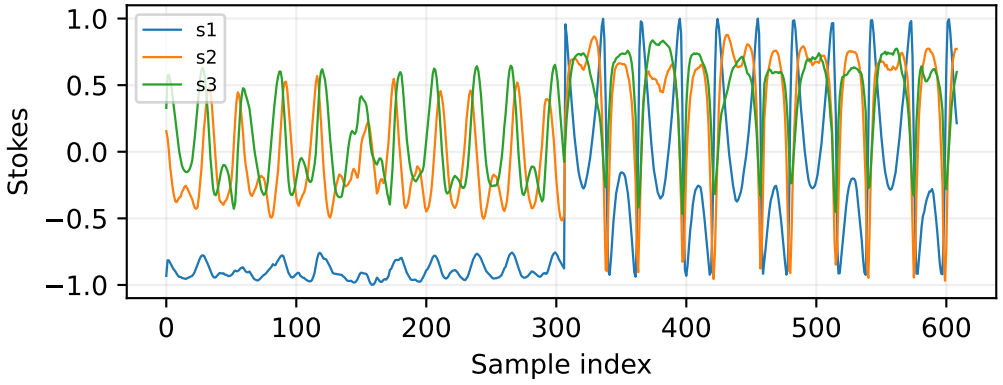


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.181, -0.494, 0.850]
Plane residual RMS: 0.09624
Rising/falling angular RMS: 34.2°
Rising/falling power RMS: 0.000121 PAX units
Cycle angular RMS: 18.8°
Cycle power RMS: 6.44e-05 PAX units
Cycle DoP RMS: 0.0325
Tracking RMS: 0.02953 V
Reference coverage: 97.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.15, -0.60, 0.79], residual 0.0702
Cycle 1 plane: [0.22, -0.41, 0.89], residual 0.054
Cycle 2 plane: [0.18, -0.49, 0.85], residual 0.0572
Cycle 3 plane: [0.16, -0.49, 0.86], residual 0.0825

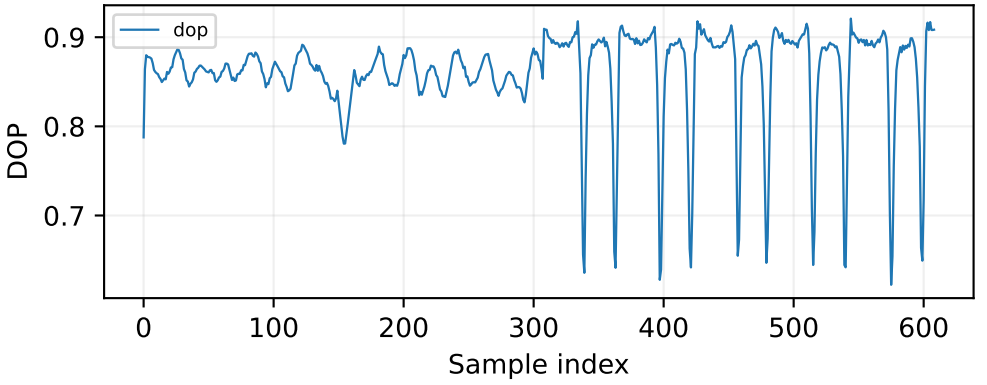
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

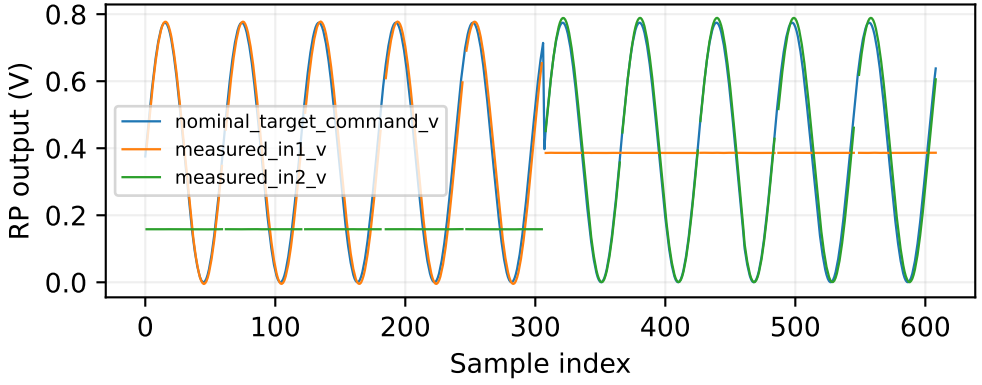
Normalized Stokes



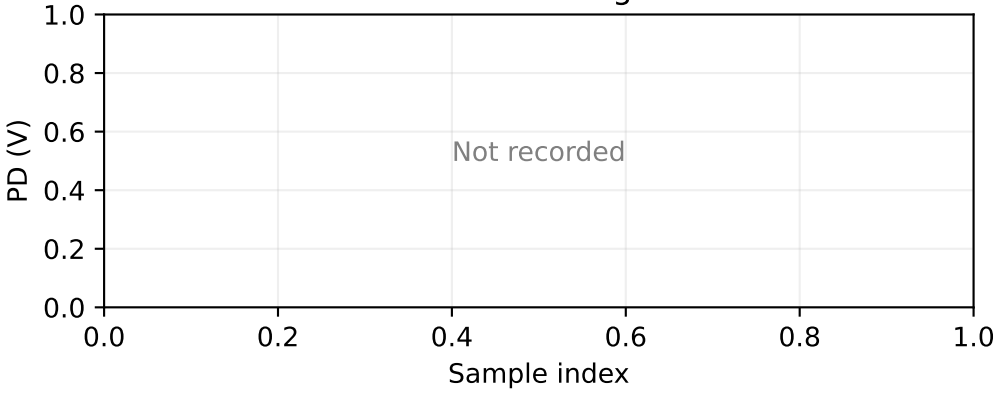
Raw degree of polarization



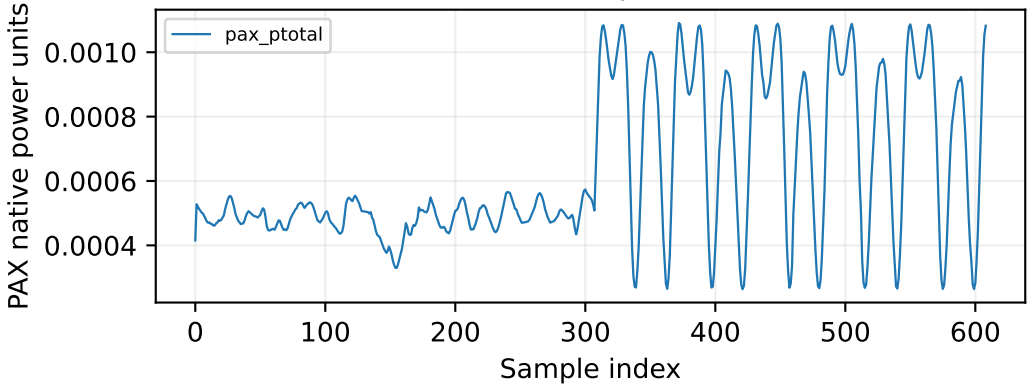
Actuator command / reference history



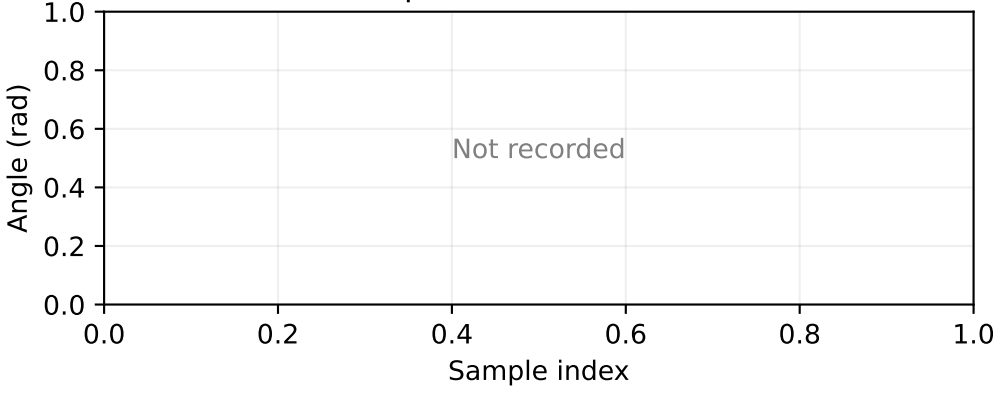
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"timing-calibration-post-in2-lv"
comment	"End-to-end measurement-tool calibration after IN2 LV jumper re
actuator	"both"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	1
cross_sweep_recorded_cycles	4
cross_sweep_phi1_bias_voltages	[0.3875]
cross_sweep_phi2_bias_voltages	[0.155]
cross_sweep_settle_s	2.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0
cross_sweep_pax_latency_s	0.12